Monday - take-home for Test 2 is due

No ~~office~~ student hours next Thurs.

Go to the MLC & work with class mates!

Read the text book

Next Friday Jon will be guest lecturing.

§8.2 IVP or Cauchy Problem

Initial value problem

Find $u \in C^2(\mathbb{R}^{n+1})$ so that

$$u_{x_1x_1} + \dots + u_{x_nx_n} - u_{tt} = 0 \qquad \underline{x} \in \mathbb{R}^n, \quad t \in \mathbb{R}$$

$$\left.\begin{aligned} u(\underline{x},0) &= \phi(\underline{x}) \qquad \underline{x} \in \mathbb{R}^n \\ u_t(\underline{x},0) &= \psi(\underline{x}) \qquad \underline{x} \in \mathbb{R}^n \end{aligned}\right\} \text{ initial data}$$

Note: if $t' = -t$ $\quad u_{t'} = \frac{\partial u}{\partial t}\frac{\partial t}{\partial t'} = -u_t$

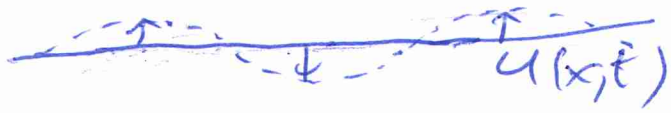
$u_{t't'} = \qquad = +u_{tt}$

the wave eq'n doesn't change. So we only need to get the soln for $t>0$ (to get the sol'n for $t<0$, replace t by $-t$ in the sol'n).

Physical interp. of IVP

$n=1$ $\quad u_{xx} - u_{tt} = 0$ $\quad$ ∞-long (guitar) string.

u is the displacement from equil.



$u(x,0) = \phi(x)$ initial ~~pos~~ displacement

$u_t(x,0) = \psi(x)$ initial velocity

$n=2$ $\quad u_{xx} + u_{yy} - u_{tt} = 0$ $\quad$ displacement of a membrane (drum - ∞ radius)

$n=3$ $\quad u_{xx} + u_{yy} + u_{zz} - u_{tt} = 0$ $\quad$ acoustic waves (sound through air, water)

well-posedness for $n=1$

$$u_{xx} - u_{tt} = 0 \quad x \in \mathbb{R},\ t > 0$$

$$\left.\begin{aligned} u(x,0) &= \phi(x) \\ u_t(x,0) &= \psi(x) \end{aligned}\right\} x \in \mathbb{R}$$

solution:

$$u(x,t) = F(x-t) + G(x+t) \quad \text{(general sol'n)}$$

$$= \frac{\phi(x+t) + \phi(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} \psi(x')\,dx'$$

(d'Alembert sol'n)

Gives us existence.

Uniqueness? Assume 2 sol'ns: $u_1(x,t)$, $u_2(x,t)$

$\tilde{u} = u_1 - u_2$ To show $\tilde{u} = 0$

$\tilde{u}_{xx} - \tilde{u}_{tt} = 0$ $\quad \tilde{u}(x,0) = 0$ $\quad \tilde{u}_t(x,0) = 0$

$\tilde{u}(x,t) = F(x-t) + G(x+t)$

(1) $\tilde{u}(x,0) = F(x) + G(x) = 0$

$\tilde{u}_t(x,t) = F'(x-t)(-1) + G'(x+t)(+1) = 0$

(2) $\tilde{u}_t(x,0) = -F'(x) + G'(x) = 0$

$\cancel{F'(x) = G'}(x)$

From (1) $F(x) = -G(x) \Rightarrow F'(x) = -G'(x)$

subst. into (2) $G'(x) + G'(x) = 0 \Rightarrow G'(x) = 0$

$\Rightarrow G(x) = c_1$ subst. into (2) $\Rightarrow F(x) = c_2$

subst. into (1) $c_1 + c_2 = 0 \Rightarrow c_1 = -c_2$

$\Rightarrow \tilde{u}(x,t) = F(x-t) + G(x+t)$

$= \cancel{c_1 + c_2 = 0}, \; c_2 + c_1$

$+ c_2 \cancel{+} c_2 = 0$

<u>uniqueness</u>.

d'Alembert sol'n is <u>the</u> sol'n

contin. dep. on data: follows from d'Alembert sol'n.

Graphs of sol'ns

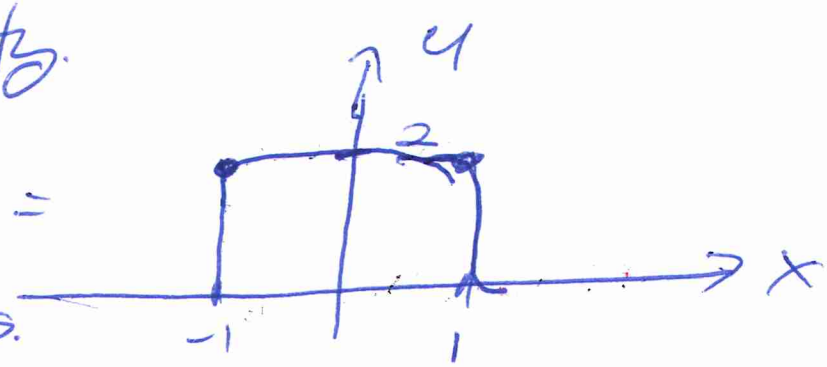
$n=1$ $\quad u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(x')\,dx'$

$-\infty < x < \infty, \quad t \geq 0$

$\phi(x)$ is initial displacement

$\psi(x)$ " " velocity.

ex) $\psi(x) = 0$ $\qquad \phi(x) =$



$t=0$

Graph the sol'n for different times.

$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right]$

$\underline{t = 1/2}$

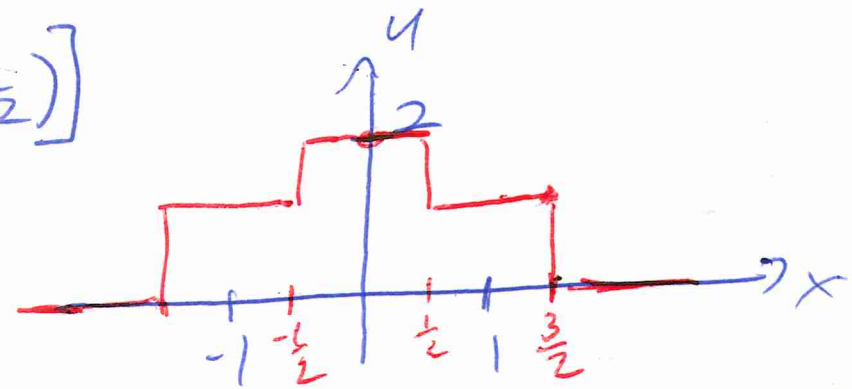
$u(x,t) = \frac{1}{2}\left[\phi(x+\frac{1}{2}) + \phi(x-\frac{1}{2})\right]$

$x=0 \qquad \frac{1}{2}\left[\phi(\frac{1}{2}) + \phi(-\frac{1}{2})\right]$

$= \frac{1}{2}(2+2) = 2$

$x = \frac{1}{2} \qquad = \frac{1}{2}\left[\phi(1) + \phi(0)\right] = 2$

$x = 3/4 \qquad = \frac{1}{2}\left[\phi(\frac{5}{4}) + \phi(\frac{1}{4})\right] = 1$



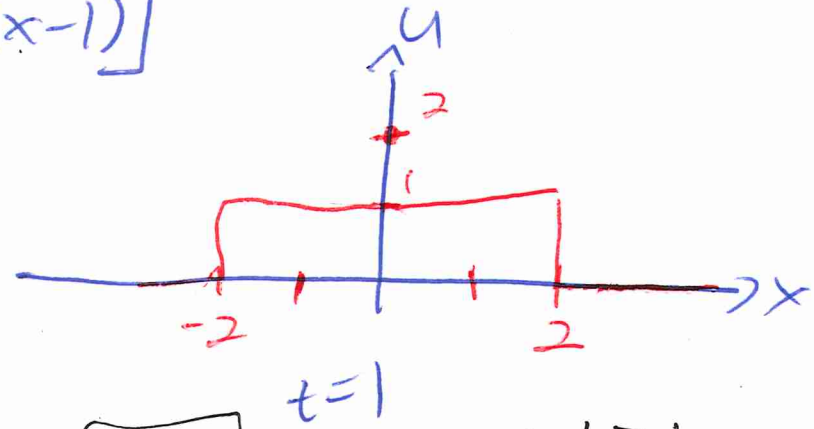
$t = 1/2$

$\underline{t=1}$

$$u(x,t) = \tfrac{1}{2}\left[\phi(x+1) + \phi(x-1)\right]$$

$x=0 \quad \tfrac{1}{2}\left\{\phi(1) + \phi(0)\right] = 2$

$x = \tfrac{1}{4} \quad \tfrac{1}{2}\left[0 + 2\right) = 1$



$u=0$ when $x+\frac{1}{2} > 1 \Rightarrow \boxed{x > \frac{1}{2}}$

and $x - \frac{1}{2} > 1 \Rightarrow x > 3/2$

$x+1>1$ or $x+1<-1$

and $x-1>1$ or $x-1<-1$

$(t=1)$

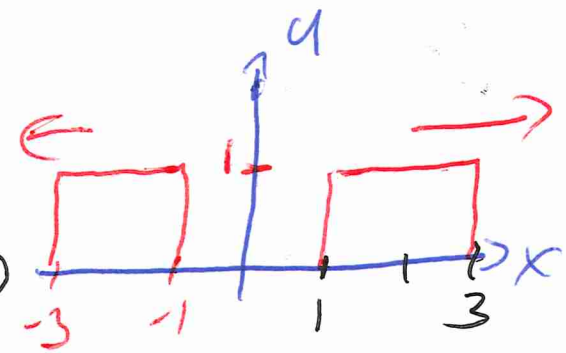
$x+1>1$

and $x-1>1$

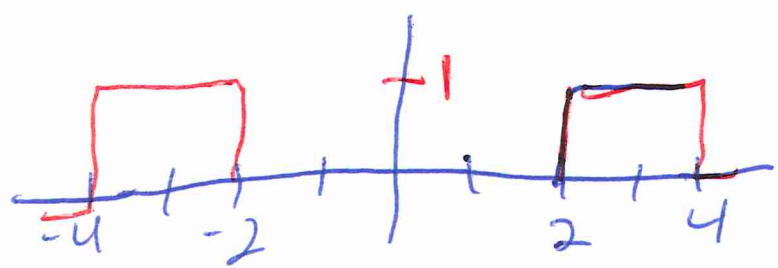
$x>0$ and $\boxed{x>2}$

$t=2 \quad u(x,t) = \tfrac{1}{2}\left[\phi(x+2) + \phi(x-2)\right]$

$x=0 \quad = \tfrac{1}{2}\left[\phi(2) + \phi(-2)\right] = 0$



$t=2$



$t=3$